

Welcome! 🍌

EECS 376, Sec. 18

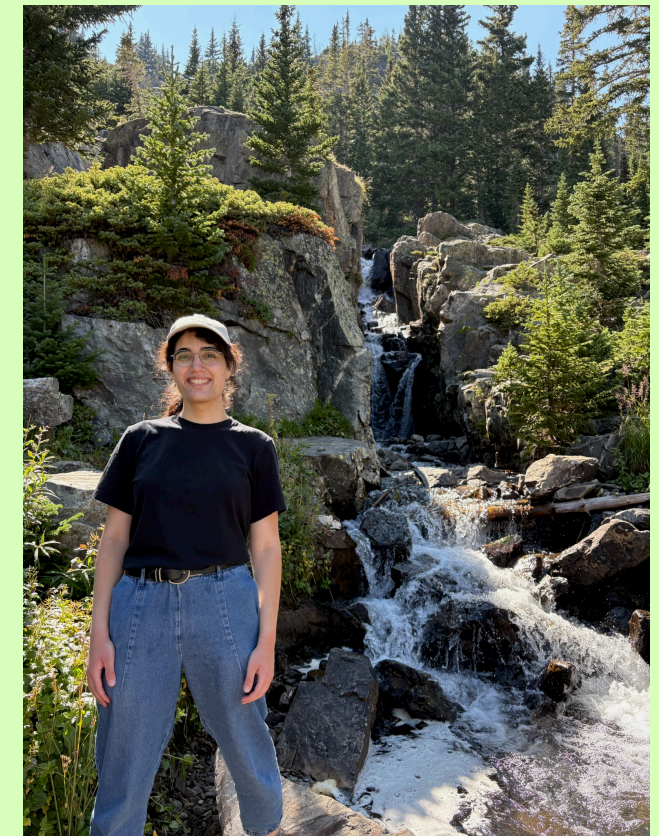
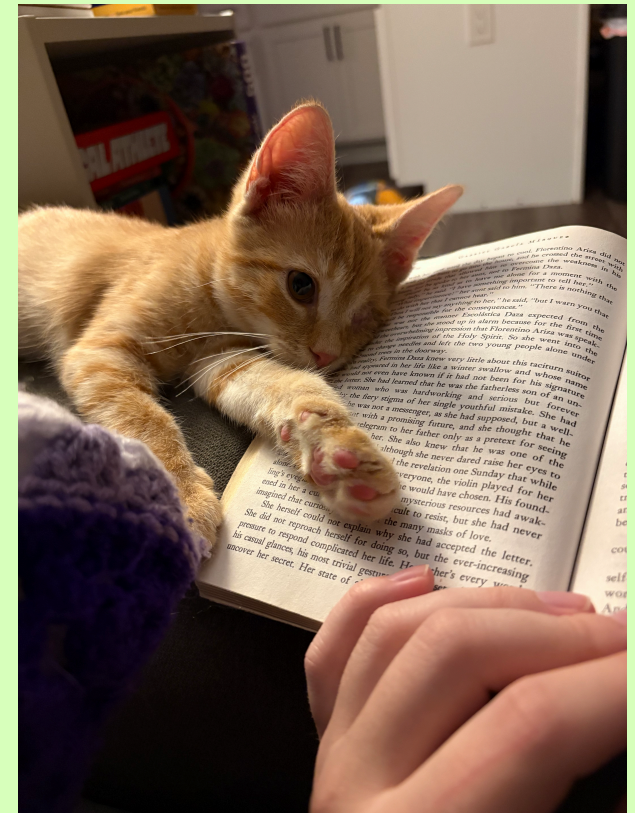
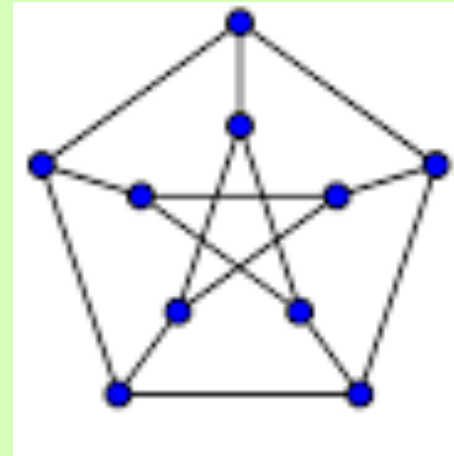
D1: Introduction, proof techniques, asymptotic notation, potential method review

Agenda

- I yap at you for 5-10 min
 - Intro to the course
 - Quick review of material this week
- Worksheet (remaining time)

About me

- I'm Sam Hiken (she/her)
- 2nd-year PhD student
- First-time GSI
- Talk to me about: graph algorithms, cooking, hiking, crossword puzzles, my roommate's new cat
- **Office hours:** tentatively...
 - Mon 10:30-12:30, Leinweber 2421
 - Tue: 2:30-4:30, Leinweber 2421
 - Wed: 10:30-12:30, Leinweber 2421, 2:30-4:30, central campus location TBD



Discussion attendance policy

- Discussion attendance is not mandatory. However...
- If you attend at least 10/14 discussions throughout the semester:
 - You get 3% for free added to your overall grade, and your exam scores weigh 3% less
- You can mark your attendance today by scanning a QR code on the last slide

Two more things:

1. Take care of yourself

- Drink lots of water, sleep, eat healthy, exercise
- Start assignments early
- Remember: the stress is temporary

2. You will learn the most from the homework

- Struggling with hard problems is how we learn
- Don't use LLMs
 - If you can do the homework problems, you can do the exam problems

Now for the review...

Review: proof methods

- **Direct proof:** Show $P \implies Q$ via a sequence of logical statements, each of which follows directly from the last
- **Proof by contrapositive:** Show $P \implies Q$ by showing that

$$\neg Q \implies \neg P$$

- Ex: “*If an object is orange, it has a color*” is equivalent to saying, “*If an object does not have a color, it is not orange*”
- **Proof by contradiction:** Show $P \implies Q$ by showing that $\neg(P \implies Q)$ implies something that cannot be true



Review: induction

We use induction when we want to show that some property P holds for all integers greater than or equal to some integer j .

An inductive argument has two parts:

1. **Base case:** We show that j has property P .
2. **Inductive step:** We show that, for any $k > j$:

if $j-1$ has P , then j has P

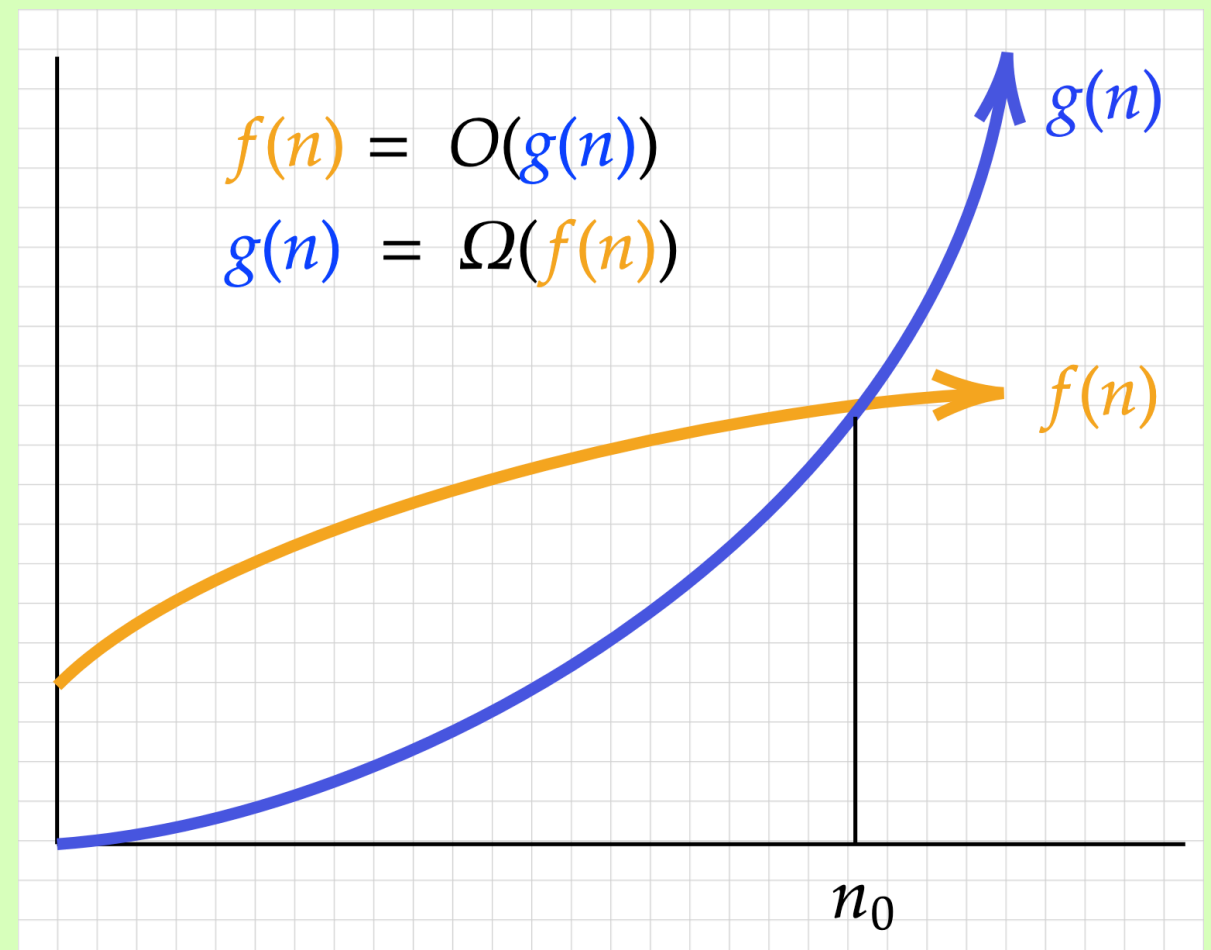
Review: asymptotic notation

Big-O is an upper-bound, Big- Ω is a lower-bound, Big- Θ is both:

Definition 1. We say that $f(n) = O(g(n))$ if there exist positive constants $c, n_0 > 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$.

Definition 2. We say that $f(n) = \Omega(g(n))$ if there exist positive constants $c, n_0 > 0$ such that $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$.

Definition 3. We say that $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.



Review: potential method

Useful tool for bounding the runtime of an algorithm:

1. Define a “potential function” s
2. Show that:
 - $s = B$ at the start of the algorithm.
 - Each step of the algorithm decreases s by 1.
 - When $s = 0$, the algorithm terminates.

Then the algorithm can only take B steps.

Attendance QR code

